Michał Siniarski

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Personal Profile

AI Engineer and Warsaw University of Technology graduate with practical experience in deep learning, computer vision, and multi-modal signal processing for aerospace and defense applications.

Experience

Air Force Institute of Technology | AI Engineer
Warsaw, Poland
July 2024 - Present

- Planning and coordinating work for the AI team: assigning tasks to team members, setting priorities, and tracking deliverables,
- Researching and prototyping machine learning models for experimental applications,
- Presenting results to clients and adapting models to their specific needs,
- Automating research and development workflows with AI coding agents (Claude Code, Codex).

Key achievements and projects:

- CNN and ResNet based models for frequency domain signals (FFT) and spectrograms (STFT) analysis, SAE model for sound denoising,
- RandLANet model and pipeline implementation for large scale, sparse point cloud semantic segmentation,
- Automated hyperparameter optimization system built with Optuna,

Air Force Institute of Technology | Project Manager
Warsaw, Poland,
April 2023 - June 2024

Led team activities within the European HYDEF defense program, coordinating direct-hit warhead evaluation and consortium deliverables.

Core contributions:

- Planning FEM simulations for structural and design evaluation,
- Managing project delivery and governance in accordance with PRINCE2,
- Leading assigned Work Packages within the international consortium,
- Presenting technical work plans and results to international partners,

Air Force Institute of Technology | Internship
Warsaw, Poland,
July 2022

Developed a Python-based Non-Destructive Testing (NDT) system for defect detection in aerospace components, covering data acquisition, signal processing, and visualization.

Education

MSc **Warsaw University of Technology, Faculty of Power and Aeronautical Engineering**: 2023 - 2025
Aeronautical Engineering

- Specialization: Automation and Aerospace Systems,
- Master's Thesis: Structural Health Monitoring (SHM) system using ML for direct-hit parameter estimation,
- Graduated with distinction.

Eng **Warsaw University of Technology, Faculty of Power and Aeronautical Engineering**: 2019 - 2023
Aeronautical Engineering

- Specialization: Aircraft Propulsion,
- Engineering Thesis: Automated NDT inspection station for aircraft components, including hardware selection and open-source software implementation.

Publications

Development of an Open-Source Robotic NDT Solution for Automated Composite Repair Testing Oct 2024

Michał Siniarski, Piotr Synaszko, Krzysztof Dragan

doi.org/10.2478/fas-2023-0009

Skills and technologies

Tech stack:

- **Core & ML**: Python, PyTorch, TensorFlow, scikit-learn, OpenCV, NumPy, pandas, Optuna,
- **Languages**: Rust (basic), C (basic), SQL, MATLAB, LaTeX,
- **Developer Tools**: Docker (basic), Git / GitHub, Conda, UV, Linux,

Spoken Languages:

- Polish (native),
- English (fluent).

Interests

- PC building,
- Photography.

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